Daksh Daksh (daksh-daksh.github.io)

Postdoctoral Researcher, IMS SAS, Bratislava, Slovakia

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Computer science researcher skilled in supervised, unsupervised, and reinforcement learning, including generative and multimodal AI. Focused on scalable, automated machine and deep learning for scientific applications.

Research Experience

Postdoctoral Researcher, Institute of Measurement Science (IMS), Department of Imaging Methods, Slovak Academy of Sciences (SAS), Bratislava, Slovakia Sep 2026 – Present

- Working on Biomedical image datasets of MRI using deep learning.
- Developing an automated toolkit for the optimization, segmentation, and classification of multi-label objects.

PhD Researcher & affiliated with CRC 1066, MPIP Mainz Sep 2021 – March 2026

- Working on automated image registration for correlative microscopy using deep learning.
- Developed Python tool for training free multi-modal image registration of light and electron microscopy and supervised image preprocessing tool for generating the precised image-pair.
- Worked on deep Learning for image analysis and microscopy data analysis using Machine Learning e.g. for generation of super resolution microscopy images

Web group Coordinator, Max Planck PhDnet, MPG 01/2023 – 01/2025

- Development and maintenance of the PhDnet website and mailing lists.

Research fellow in CSIE & IEO dept., NTNU, Taiwan 05/2018 – 07/2020

Jointly worked on project with NVIDIA AI Tech, Taiwan

- Worked on optical defect detection using supervised deep learning for Digital Holography.
- Worked on data Augmentation using GAN i.e. AC-GAN and DC-GAN and for Deep Learning Based on Digital Holograms.
- Worked on deep Learning for image analysis and generating different holographic image datasets.

Scientific Co-worker, University of Greifswald, Germany 06/2017 – 02/2018

- Worked on computational modeling and simulations for bio-molecules.

Setup configuration specialist, Aon Hewitt Pvt. Ltd., India (got selected among 5000 students & even before final semester) 11/2012 – 07/2014

- Worked on IBM mainframes, Lotus notes, C++, ASP .net (C#), VB, and SQL

EDUCATION

Max-Planck Institute for Polymer Research (MPIP) & Johannes Gutenberg University Mainz (JGU), Mainz, Germany
PhD - Computer Science (Interdisciplinary) Oct 2021 – Feb 2026
Advisor: Prof. Dr. Katharina Landfester & Dr. Ingo Lieberwirth

National Forensic Sciences University (AIR – 159 in Computer Sciences), Gandhinagar, India
MS - Forensic Nanotechnology (Nano-engineering) Jul 2014 – May 2016
Graduated in First class with Distinction

Lovely Professional University, Phagwara, India
B.Tech (Honors) - Computer Science & Engineering Jul 2008 – May 2012
Graduated in First class with Distinction

Patent

Multimodal Training-Free Registration Using Mutual Image Information. Application submitted for patent & commercialization with Max Planck Innovations 2025.

Conference Talks

Multi-resolution Cross-modality Image Registration Using Unsupervised Deep Learning Approach. D Daksh, A Kaltbeitzel, K Landfester, I Lieberwirth. Microscopy and Microanalysis: Microscopy Society of America 2023.

Unsupervised Deep Learning approach for image registration in Correlative Microscopy for the localization of Nanoparticles. D Daksh, A Kaltbeitzel, G Glaßer, I Lieberwirth, K Landfester. Invited Speaker, BIO Web of Conferences, European Microscopy Congress 2024.

Correlative Microscopy Strategies for the Identification of Intracellular Nanoparticles and their Cellular Processing. I Lieberwirth, S Han, A Kaltbeitzel, G Glaßer, D Daksh, K Landfester. Microscopy and Microanalysis: Microscopy Society of America 2024.

Publications

TFUDL-CLEM: A Training-Free Unsupervised Deep Learning Registration Method for Correlative Light and Electron Microscopy. D. Daksh, A. Kaltbeitzel, G. Glaßer, K. Landfester, I. Lieberwirth. Medical Image Analysis (Under peer review)

A Weakly Supervised Pre-processing Pipeline for Multi-Modal Image Pair Generation for Image Registration. D. Daksh, A. Kaltbeitzel, G. Glaßer,K. Landfester, I. Lieberwirth. Intelligent Systems with Applications (Under peer review)

Symposium Presentations & Talks

Microscopy data analysis: machine learning and the BioImage Archive. EMBL Heidelberg, Germany May 2022

First EMBL Imaging centre Symposium: Enabling imaging across scales. EMBL Heidelberg, Germany May 2022
Oral short talk

SFB 1066 and 1278 Joint Summer school. Fulda, Germany July 2022
Poster Presentation

Joint Symposium of RMaP and SFB 1066. Mainz, Germany July 2022
Poster Presentation

EMBO workshop: from Cryo-EM to multi-scale modelling. EMBL Heidelberg, Germany February 2023
Poster Presentation

Deep Learning & computer vision (DLCV) school. Genova, Italy June 2023
Selected among 1% with full scholarship to attend. Won the first prize for the most novel idea and best presentation.

Joint Symposium of SFB1066 and TRR319. Mainz, Germany April 2024
Poster Presentation

Internal Symposium of SFB 1066. Mainz, Germany July 2024
Poster Presentation

International Symposium of SFB 1066. Mainz, Germany January 2025
Poster Presentation

Certificates

Optimizing writing strategies. MPIP Mainz, Germany feb. 2024
German B1. In-house course in MPIP Mainz, Germany Sept. 2024
Python for HPC. MPCDF Garching, Germany Nov. 2024
Parallel Computing With Matlab. MPCDF Garching, Germany Dec. 2024
Workshop on AI and Research in MPG. Berlin, Germany Dec. 2024

Skills

Programming Languages & Tools: Python, PyTorch, Pandas, Cuda, Anaconda navigator with different IDEs, Matlab, Napari, ImageJ, Bash, Git, LaTeX, Keras, Automated ML/DL pipelines, TensorFlow, Scikit-learn, Scipy, Numpy, C, C++, C#, .Net, SQL, Database Managemnet system, HTML, PHP, IBM mainframes, Lotus notes, LLMs
Operating systems: Windows, Linux, MacOS
Languages: English (Fluent), Hindi (Native), German (B1/B2), French (Basic)